جامعه طرابلس | كليه طب الأسنان

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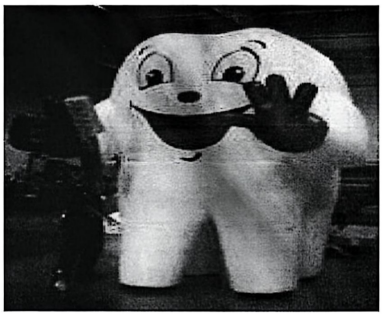
Osama

College of dentistry

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مصور كلية الأسنان
(ابن المنظور)



TOOTH DEVELOPMENT AND GROWTH

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TOOTH DEVELOPMENT (ODONTOGENESIS)

- Tooth development start at 5th week of intrauterine life (W.I.U.L) and takes place in many morphologic stages.

Morphological stages of tooth development

- The developmental stages is based on the shape of the developing tooth, which are:

1. Dental lamina.

2. Bud – stage.

3. Cap – stage.

4. Bell – stage (early and late).

(1) Dental lamina:

- During 6th (W.I.U.L), in the region of the future alveolar processes, the underling ectomesenchyme cells induce the oral epithelium to proliferate and invaginate into the mesenchyme to form the **primary epithelial bands.**
- By the 7th week the **primary epithelial bands** divides into two laminae:

a) a buccally located **dental lamina**(will give teeth)

b) alingually situated **vestibular lamina** (will form vestibule)

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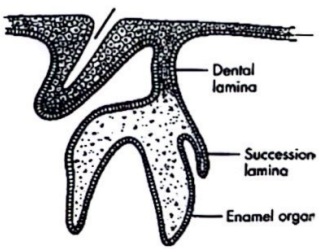
a) dental lamina

- dental lamina is formed by proliferation of the basal layer of the oral epithelium in the underlying ectomesenchyme.
- **Neural crest cells → Induction → Proliferation of basal cell layer of oral epithelium → Dental lamina**



b)The vestibular lamina:

- The vestibular lamina arises from ectodermal thickening that proliferate into the underlying ectomesenchyme later to dental lamina



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(2) Bud stage:

- labial cells of dental lamina proliferate to give bud- shaped round or ovoid epithelial structure called **dental or enamel organ** (10) in each arch.
- The ectomesenchymal cells condense beneath the tooth bud to form **dental papillae** that encapsulating the dental organ is called **dental sac** (tooth follicle).
- The dental organ, dental papilla and dental sac together constitute the **tooth germ**.

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(3) Cap stage:

1. Dental lamina: short and broad.

- The bud grows unequally to give concave free end to form cap shape.

2. Dental organ:

- Basement membrane is continuous around it separating from the sac, papillae.

a) Outer dental(enamel) epithelial:

- Single layer of cuboidal epithelial cells.
- On outer convex surface.

b) Inner dental(enamel) epithelial:

- Single layer of columnar epithelial cells.
- On concave surface.

c) Stellate reticulum:

- several layers of star-shaped cells occupying the central portion.
- connected to each other, to outer, to inner epithelial by desmosomes.

Two structure in dental organ in cap stage:

i) Enamel knot:

- localized cellular condensation at the center of inner epi.

ii) Enamel cord:

- Cellular strand running from the knot to outer epi.

* There is area of **cell free zone** between dental papilla and dental organ.

3. dental papilla:

- Ectomesenchymal condensation beneath the IEE, more cellular.

4. dental sac:

- Ectomesenchymal condensation around organ, papilla more fibrous.

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(4) Bell stage:

- The unequal growth continuous to give bell **shape.**

❖ **The early Bell stage:**

1. Dental lamina:

- The main dental lamina break up by mesenchymal invasion into:

a) Lateral dental lamina that that carrying the deciduous tooth.

b)The dental lamina proper (Successional) forming the permanent successors.

2. Enamel organ:

a) Outer dental epithelium:

- become low cuboidal.

b) Inner dental epithelium:

- These cells show elongate toward papilla and closing the cell free zone.
- The cells of inner dental epithelial inducing cells of dental papilla to differentiate into odontoblasts.

c) Stellate reticulum:

- More intercellular fluid.

d) Stratum intermedium:

- It is composed of 2 – 3 layers of flat cells between the inner dental epithelium and stellate reticulum.
- contain the **enzyme alkaline phosphatase** which is essential for mineralization of enamel.

3. Dental papilla:

- Ability to produce dentin.

4. Dental sac:

- Outer circular fibres, inner vascular layer.

✱ ***Cervical loop:*** The inner and outer enamel epithelium meets together.

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Early bell stage

❖ **The Late Bell stage:**

1. Dental lamina:

- mesenchymal invasion disintegrates dental lamina to give epithelial **rests of Serres (Serres' pearls).**

2. Dental organ:

- The enamel organ shows an increase in size in the area apart from matrix deposition.

a) Outer dental epithelium:

- first layer of dentine formed cuts nutrition to dental organ through inner epithelial.

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- Outer epithelial develop microvilli and folds on their outer surface to increasing surface area and sac form capillary loop for nutrition.

b) Inner dental epithelium:

- Dental papilla induce inner enamel epithelial to become active cell and differentiate to ameloblast to form enamel(reciprocal induction).

c) Stellate reticulum:

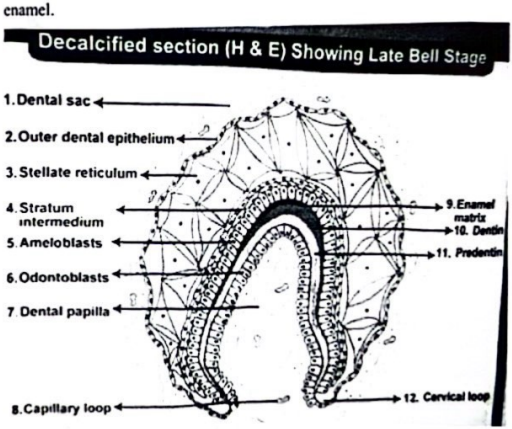
- Shrinks by loss of intercellular fluid providing :

  1-space for developing enamel.

  2-less distance between inner, outer epi. For nutrition.

d) Stratum intermedium:

- Contain alkaline phosphatase enzyme for the mineralization of the enamel.



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A. Histolophysiological Stages

- Except for initiation, there is an overlap off all stages but each tends to predominate in certain period.

1. Initiation of Dental Lamina

- D. Lamina & tooth bud is the representative of this stage.
- Lack of initiation may result in absence of one or more teeth (anodotia).
- Abnormally high initiation result in formation of super numerary teeth or tooth.

2. Proliferation

- It's the active stage.
- Proliferative growth cause regular change in size & proportion of the growing tooth.
- It's found in D. lamina, bud, cap and early bell stages.

3. Histodifferentation

- Occurs when the stratum intermedium, odontoblasts and ameloblasts are differentiated.
- Represented by the early and advanced bell stage.

4. Morphodifferentation

- Morphological pattern is determined by differential growth.
- Starts from early bell stage.
- Established by out lining the future dentine enamel junction & cemento enamel junction.

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- Disturbance in this stage affect in the form & size of the tooth.

5. Apposition

- Present in late bell stage by deposition of newly synthesized hard tooth structure & their mineralization.
- Presence of periods of activity & rest result in formation of Incremental lines in all hard tissues.

Root formation

(1) Root formation start when enamel, dentine reaches the future cervical line.

(2) The cervical loop proliferate to form **Hertwig's epithelial root sheath (HERS).**

(3) horizontal proliferate of cervical loop to form **epithelial diaphragm** ⟶ primary closer to apical foramen ⟶ .{ **soft** }.

(4) The cervical loop proliferate vertically :

a) inner enamel epithelium induce unmesnchym cells of dental papilla to differentiate into odontoblast to form root dentin.

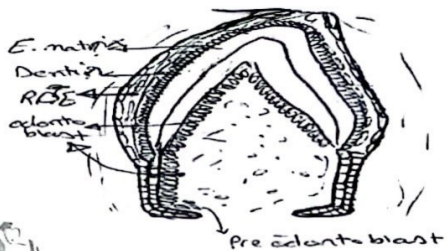
c) After first layer of dentin formation (UMCs) of dental sac degeneration of inner, outer enamel epithelium and contact to dentin and form

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cementoblast cementum secondary closer to apical foramen { **hard** }.

d) (H.E.R.S) loss its continuity by involving UMCs of dental sac and leave remnant called {**epithelial rest of malassez**}.

(5) function of epithelium mold shape of the root.



Root formation.

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1. Multirooted teeth
   - Like single root.
   - E. D. grows horizontally forming 2 tongue like projections double rooted teeth and 3 tongue like projections in triple rooted teeth.
   - These projections will grow towards each other unit dividing the wide apical foramen into 2 or 3 openings.

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On the pulpal surface of the tongue like projection, dentin will be formed then the epithelium degenerate and cementum deposited on the outer surfaces of the united projection.



I- CLINICAL CONSIDERATION:

1. Enamel pearl and bare dentin:
   - When part of E. R. S. H. failure to degenerate so cells will diff. to Ameloblasts to produce enamel.
   - It's mainly happens at the furcation areas
   - sometime it will remain covering the dentin surface, thus preventing cementum to be formed leading to bare dentin.

2. Accessory root canal

   It's connection between pulp & PDL other than apical foramen due to:-

   - Deffective formation of E. R. S. H.

   - Degenerated of E. R.S. H. before odontablastic differentiation

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- Displaced B. V.

- Failure of fusion of tongue like projection of E. D.

3. Intermediate cementum

- Due to degenerate of the E.R.S.H. after odontoblastic differentiation but before they lay down dentin.

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